

EXCEPTION PRACTICE PROGRAM

Q1.

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         try {
8             System.out.println("Enter numerator: ");
9             int numerator = sc.nextInt();
10
11             System.out.println("Enter denominator: ");
12             int denominator = sc.nextInt();
13
14             int result = numerator / denominator;
15             System.out.println("Result: " + result);
16         } catch (ArithmeticException e) {
17             System.out.println("Error: Cannot divide by zero!");
18         } finally {
19             sc.close();
20         }
21     }
22 }
```

Input

```
10
2
```

Output

```
Enter numerator:
Enter denominator:
Result: 5
```

Q2.

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         int[] arr = {10, 20, 30, 40, 50};
6         Scanner sc = new Scanner(System.in);
7
8         try {
9             System.out.print("Enter an index (0-4): ");
10            int index = sc.nextInt();
11
12            System.out.println("Element at index " + index + " is: " + arr[index]);
13        } catch (ArrayIndexOutOfBoundsException e) {
14            System.out.println("Error: Index is outside the valid range");
15        } finally {
16            sc.close();
17        }
18    }
19 }
```

Run

Save

Input

2

Output

```
Enter an index (0-4): Element at
index 2 is: 30
```

Q3.

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         System.out.print("Enter a number: ");
7         String input = sc.nextLine();
8
9         try {
10            int num = Integer.parseInt(input);
11            System.out.println("Valid integer entered: " + num);
12        } catch (NumberFormatException e) {
13            System.out.println("Invalid input! Please enter a valid integer");
14        } finally {
15            sc.close();
16        }
17    }
18 }
```

Run

Save

Input

1
2

Output

```
Enter a number: Valid integer
entered: 1
```

Q4.

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both `ArithmeticException` and `ArrayIndexOutOfBoundsException` using multiple catch blocks.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int[] result = new int[1];
7
8         try {
9             System.out.print("Enter first number: ");
10            int num1 = sc.nextInt();
11            System.out.print("Enter second number: ");
12            int num2 = sc.nextInt();
13
14            int division = num1 / num2;
15
16
17            result[1] = division;
18
19            System.out.println("Result stored in array: " + result[1])
20
21        } catch (ArithmeticException e) {
22            System.out.println("ArithmeticException caught: " + e.getM
23        } catch (ArrayIndexOutOfBoundsException e) {
24            System.out.println("ArrayIndexOutOfBoundsException caught:
25        } finally {
26            sc.close();
27    }
```

Input

10
2

Output

Enter first number: Enter second
number:
ArrayIndexOutOfBoundsException
caught: Index 1 out of bounds
for length 1